

Mathematics, 2017 EC, natural science, entrance exam, round 1

Grade 12 · Mathematics · 2017 E.C.

Time allowed: 120 minutes

Questions: 50

Total marks: 50

Name _____ Grade _____ ID No. _____ Date _____

1. Which one of the following function is NOT a power function? (1)
- A. $f(x) = x^{-\frac{3}{5}}$ B. $f(x) = \sqrt{2}x^{\frac{3}{2}}$
 C. $f(x) = \pi x^5$ D. $f(x) = (\sqrt{3})^x$
-
2. Let $f(x) = (1 - 2x)^{\frac{1}{2}}$. Then what is the values of $f'(x)$? (1)
- A. $\frac{1}{\sqrt{1-2x}}$ B. $-\frac{1}{2}\sqrt{1-2x}$
 C. $\frac{1}{2}\sqrt{1-2x}$ D. $-\frac{1}{\sqrt{1-2x}}$
-
3. Which of the following statement describes the difference between scalar and vector quantities? (1)
- A. Physical quantities that only have direction are scalars, while that have only magnitude are vectors.
 B. Physical quantities that only have magnitude are scalars, while that have both magnitude and direction are vectors.
 C. Physical quantities that only have magnitude are vectors, while that have both magnitude and direction are scalars.
 D. Physical quantities that only have direction are vectors, while that have both magnitude and direction are scalars.
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4. Consider the following data: 1, 3, 7, 9, 11, 13, 18, 28. Then the coefficient of mean deviation about the median is equal to: (1)
- A. $\frac{1}{4}$ B. $\frac{8}{5}$
 C. $\frac{5}{8}$ D. $\frac{1}{2}$
-
5. Which one of the following is an irrational number? (1)
- A. 3.4000 B. $\sqrt{0.04}$
 C. 1.16222... D. $\sqrt{29}$
-
6. Which one of the following statements defines favorable outcomes in a random experiment? (1)
- A. The outcomes that insure the occurrence of a particular result.
 B. The outcomes containing all sample elements.
 C. The outcomes containing only one sample element.
 D. The outcomes that have equal chance of being chosen.
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7. Given that $\log 4.8 = 0.6812$, $\log 6.93 = 0.8406$ and $\log 6.84 = 0.8351$, then what is the value of $\sqrt{48}$? (1)
- A. 6.83 B. 6.71
 C. 6.93 D. 6.84

8. Let $f(x) = x^3 + 2x + 1$ and $g(x) = x^2 + 1$ be given two polynomial functions. Then which of the following is true? (1)
- A. $\frac{f(x)}{g(x)} = \frac{x^2+1}{x-1}$ B. $f(x) + g(x) = x^3 + x^2$
 C. $f(x) - g(x) = x^3 - x^2 + 2x$ D. $f(x)g(x) = x^5 + x^3 + x^2 - 2x - 1$
-
9. Let $A = \{x | 1 \leq x \leq 10\}$ and $B = \{x | -1 \leq x \leq 10\}$ be two sets of real numbers. What is the relative complement of A with respect to B ? (1)
- A. $\{x | -1 \leq x \leq 0\}$ B. $\emptyset$
 C. $\{x | -1 \leq x \leq 1\}$ D. $\{x | -1 \leq x < 1\}$
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10. Consider the function $f(x) = x^3 - 3x^2 + 3$ on $(-2, 3)$. Then which of the following point is local minimum and local maximum of $f(x)$ respectively? (1)
- A. $(-2, -27)$ and $(0, 3)$ B. $(0, 3)$ and $(2, -1)$
 C. $(3, 3)$ and $(-2, -27)$ D. $(2, -1)$ and $(0, 3)$
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11. The table below shows a grouped frequency distribution. X 0–9 10–19 20–29 30–39 40–49 f 4 3 6 5 2 What is the class midpoint of the fourth class? (1)
- A. 34 B. 34.5
 C. 35 D. 39
-
12. Given the data set: 29, 44, 46, 52, 35, 35, 25, 15, 10, 55, 56 What is the coefficient of quartile deviation? (1)
- A. 0.25 B. 0.35
 C. 0.45 D. 0.55
-
13. Which of the following is the simplified form of the expression $\frac{81^{1/5}}{27^{3/5}}$? (1)
- A. 3 B. $\sqrt[5]{9}$
 C. $\sqrt[5]{3}$ D. $\frac{1}{3}$
-
14. What is the total surface area (in cm^2) of the cylinder whose radius is 2 cm and height is 7 cm? (1)
- A. 49π B. 28π
 C. 36π D. 98π
-
15. Suppose that we have 4 objects chosen 3 at a time. Then which of the following statement is true? (1)
- A. The number of combinations is 6. B. The number of combinations is 24.
 C. The number of permutations is 24. D. The number of permutations is 6.
-
16. Consider the equation $\log x + \log(x - 3) = 1$. Then what is its solution set? (1)
- A. $\{5\}$ B. $\{2, -5\}$
 C. $\{-2, 5\}$ D. $\{6\}$
-
17. If a fair six-sided die is tossed once, then what is the probability of getting a factor of 4? (1)
- A. $\frac{1}{2}$ B. $\frac{1}{6}$
 C. $\frac{5}{6}$ D. $\frac{1}{3}$

18. Which of the following statement is true about a relation? (1)
- A. The domain and the range of a relation are equal sets.
B. The range of a relation is the set of all its first coordinates.
C. The domain of a relation is the set of all its second coordinates.
D. A relation from A to B is any subset of $A \times B$ for non-empty sets A and B.

19. Which one of the following expression is the correct decomposition of the rational expression $\frac{2x-3}{x^2+x-2}$? (1)
- A. $\frac{5}{3x+2} + \frac{1}{x+1}$ B. $\frac{1}{3-3x} + \frac{7}{3x+6}$
C. $\frac{1}{1-x} + \frac{5}{3x+6}$ D. $\frac{7}{3x+2} - \frac{1}{x+1}$

20. What is the average rate of change of $g(x) = x^3 - 2x$ over the interval $1 \leq x \leq 6$? (1)
- A. $\frac{203}{5}$ B. 41
C. $\frac{37}{5}$ D. 46

21. Which of the following matrix is the inverse of the matrix $\begin{pmatrix} 1 & -1 & 2 \\ 1 & 0 & 1 \\ 3 & 1 & 3 \end{pmatrix}$? (1)
- A. $\begin{pmatrix} 1 & 5 & -1 \\ -3 & 0 & 1 \\ -4 & -1 & 1 \end{pmatrix}$ B. $\begin{pmatrix} -1 & 0 & 1 \\ 5 & -3 & -4 \\ -1 & 1 & 1 \end{pmatrix}$
C. $\begin{pmatrix} -1 & 0 & 1 \\ 5 & 3 & 4 \\ 1 & -1 & 1 \end{pmatrix}$ D. $\begin{pmatrix} -1 & 5 & -1 \\ 0 & -3 & 1 \\ 1 & -4 & 1 \end{pmatrix}$

22. When the number $43 \times 10^8 \times 10^{-5}$ is written in standard form, it equals (1)
- A. 4.3×10^4 B. 43×10^3
C. 4.3×10^3 D. 0.43×10^5

23. If a square pyramid has a base side of 6 cm long and its lateral faces are equilateral triangles, then what is its volume (in cm^3)? (1)
- A. 72 B. $3\sqrt{2}$
C. 48 D. $36\sqrt{2}$

24. A television company assembles flat screen and curved screen televisions. The company has to invest Birr 15,000 to assemble one flat screen and Birr 20,000 to assemble one curved screen television. The company allocates a maximum of Birr 900,000 in a week for assembling. The company gets Birr 1800 profit from a sale of one flat screen television and Birr 2000 profit from a sale of one curved screen television. Moreover, the company has a store of maximum capacity of storing 50 televisions. Let x be the number of flat screens and y be the number of curved screens. Then which one of the following inequalities should model the profit for the objective function $Z=1800x+2000y$? (1)
- A. $\begin{cases} 20,000x + 15,000y \leq 360,000 \\ x + y \leq 50 \\ x, y \geq 0 \end{cases}$ B. $\begin{cases} 15,000x + 20,000y \leq 900,000 \\ x + y \leq 50 \\ x, y \geq 0 \end{cases}$
C. $\begin{cases} 15,000x + 20,000y \leq 360,000 \\ x + y \leq 50 \\ x, y \geq 0 \end{cases}$ D. $\begin{cases} 20,000x + 15,000y \leq 900,000 \\ x + y \leq 50 \\ x, y \geq 0 \end{cases}$

25. Which one of the following distribution is multimodal? (1)
- A. 18, 14, 15, 10, 11, 3, 18, 12
B. 6, 10, 9, 6, 10, 4
C. 8, 7, 11, 13, 17, 9, 8, 7, 6, 11
D. 12, 13, 26, 15, 25, 38, 28, 27, 32
-
26. What is the equation of a line that passes through $(-2, 5)$ and has a slope $\frac{1}{2}$? (1)
- A. $2y - x - 12 = 0$
B. $y = 2x + 12$
C. $2y + x = 5$
D. $y = \frac{1}{2}x + 5$
-
27. What is the minimum value of the function defined by $f(x) = -3 + 4x + x^2$? (1)
- A. -7
B. 1
C. 0
D. -2
-
28. Which of the following statement is true about the given system of linear equations? (1)
- A. $\begin{cases} x + 3y = 6 \\ 2x - 5y = 2 \end{cases}$ is a homogeneous system.
B. $\begin{cases} 2x + y = 0 \\ x - 4y = 0 \end{cases}$ is a non-homogeneous system.
C. $\begin{cases} x + 3y = 2 \\ 2x - 5y = 1 \end{cases}$ is a non-homogeneous system.
D. $\begin{cases} 3x - y = 1 \\ x + 3y = 0 \end{cases}$ is a homogeneous system.
-
29. Suppose $f(x) = 3\sqrt{x}$ and $g(x) = \frac{1}{3}x^4 + 2x$ be given two functions. Then which of the following is equal to $(f + g)'(x)$? (1)
- A. $3x^{-\frac{1}{2}} + 4x^3 + 2$
B. $\frac{2}{3}x^{-\frac{1}{3}} + 4x^3 + 2x$
C. $\frac{2}{3}x^{-\frac{1}{3}} + 4x^3 + 2$
D. $\frac{3}{2}x^{-\frac{1}{2}} + \frac{4}{3}x^3 + 2$
-
30. The table given below shows the ages distribution of children in a small village in years. Age interval Frequency
(f) 2-4 4 5-7 1 8-10 2 11-13 3 Then what is the mean age of the children's (in years)? (1)
- A. 6.9
B. 6.6
C. 7.5
D. 7.2
-
31. A worker of a company borrowed Birr 100,000 from the company's credit and saving association. She agreed to pay 10% of the loan during the agreement and on the remaining amount to pay Birr 3000 (monthly plus interest) in 36 monthly installments. What is the total amount of money she paid to the credit and saving association? (1)
- A. Birr 106,000
B. Birr 118,000
C. Birr 108,000
D. Birr 116,000
-
32. Given the system of linear equations $\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$, where $a_1, a_2, b_1, b_2, c_1, c_2 \in \mathbb{R}$ and $a_2, b_2, c_2 \neq 0$, (1)
- then which of the following is true?
- A. If $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, then the system has one solution.
B. If $\frac{a_1}{a_2} = \frac{c_1}{c_2}$, then the system has no solution.
C. If $\frac{b_1}{b_2} = \frac{c_1}{c_2}$, then the system has unique solution.
D. If $\frac{a_1}{a_2} \neq \frac{c_1}{c_2}$, then the system has infinitely many solutions.
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33. Which of the following is true about the cosecant value of the given angles? (1)
- A. $\csc 135^\circ = -\frac{1}{\sqrt{2}}$
B. $\csc 135^\circ = -\sqrt{2}$
C. $\csc 45^\circ = \frac{1}{\sqrt{2}}$
D. $\csc 45^\circ = \sqrt{2}$

34. Which of the following statement defines geometric progression? (1)
- It is a sequence in which the ratio between consecutive terms is a non-zero constant.
 - It is a sequence in which the difference between consecutive terms is a positive constant.
 - It is a sequence in which the difference between consecutive terms is a non-zero constant.
 - It is a sequence in which the ratio between consecutive terms is a positive constant.
-
35. Consider the line segment PQ with $P = (1, -1)$ and $Q = (4, 2)$. Then which of the following point divides PQ internally in the ratio of 2:1? (1)
- $(3, 1)$
 - $(2, 0)$
 - $(2, 1)$
 - $(3, 0)$
-
36. What is the determinant of the matrix $\begin{pmatrix} \frac{1}{2} & 1 & \frac{3}{2} \\ 2 & \frac{5}{2} & 3 \\ \frac{7}{2} & 4 & \frac{9}{2} \end{pmatrix}$? (1)
- 0
 - $\frac{1}{2}$
 - $\frac{1}{8}$
 - 1
-
37. Given the objective function $Z = -x + 2y$ that is subjected to the constraints $\begin{cases} x \geq 3 \\ x + y \geq 5 \\ x + 2y \leq 6 \\ y \geq 0 \end{cases}$, what are the (1)
- maximum and minimum values of Z respectively?
- 2 and -6
 - 2 and 1
 - 1 and -2
 - 6 and -2
-
38. Let $a, b \in \mathbb{R}$ such that $a < b$ and f be a polynomial function. Then which one of the following statement states the location theorem? (1)
- If $f(a)$ and $f(b)$ have opposite signs, then there is at least one zero of f between a and b .
 - If $f(a)$ and $f(b)$ have the same signs, then there is at least one zero of f between a and b .
 - If $f(a)$ and $f(b)$ have the same signs, then there exist a unique zeros of f between a and b .
 - If $f(a)$ and $f(b)$ have opposite signs, then there is a unique zeros of f between a and b .
-
39. Suppose two triangles are similar by SSS similarity theorem. If the side lengths of one triangle are 2 cm, 4 cm and 6 cm, and the shorter side of the other triangle is 9 cm long, then what are the lengths (in cm) of the other two sides of the triangle? (1)
- 9 and 21
 - 18 and 27
 - 9 and 27
 - 18 and 21
-
40. Let $a, b \in \mathbb{R}$ such that $a < b$ and f be a polynomial function. Then which one of the following statement states the location theorem? (1)
- If $f(a)$ and $f(b)$ have opposite signs, then there is at least one zero of f between a and b .
 - If $f(a)$ and $f(b)$ have the same signs, then there is at least one zero of f between a and b .
 - If $f(a)$ and $f(b)$ have the same signs, then there exist a unique zeros of f between a and b .
 - If $f(a)$ and $f(b)$ have opposite signs, then there is a unique zeros of f between a and b .

41. If $\begin{pmatrix} x+y & 2 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 7 & 2 \\ 5 & x-y \end{pmatrix}$, then what is the value of x ? (1)
- A. -2
B. -5
C. 5
D. 2
-
42. Consider the following data: 13, 16, 19, 19, 19, 19, 24, 27 What is the mean deviation about the mode? (1)
- A. 4
B. 2.75
C. 3.75
D. 3
-
43. Which of the following statement is true about types of data: (1)
- A. The age of students in a class is a quantitative data.
B. The height of students in a class is a qualitative data.
C. The weight of students in class is a qualitative data.
D. The gender of students in a class is a quantitative data.
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44. A company offers a discount of 25% on one of its product for promotion. What is a regular price of the product if it is purchased for Birr 99? (1)
- A. 110
B. 198
C. 132
D. 125
-
45. What is the total surface area (in cm^2) of the cylinder whose radius is 2cm and height is 7cm? (1)
- A. 49π
B. 28π
C. 36π
D. 98π
-
46. Which of the following statement is true about the given infinite series? (1)
- A. $\sum_{n=1}^{\infty} (-1)^n 2^n$ is convergent.
B. $\sum_{n=1}^{\infty} \left(\frac{4}{3}\right)^n$ is convergent.
C. $\sum_{n=1}^{\infty} 2^n$ is divergent.
D. $\sum_{n=1}^{\infty} (-1)^n \left(\frac{2}{5}\right)^n$ is divergent.
-
47. Let θ be an angle in standard position and (x, y) be a point on the terminal side of θ , which is different from the origin. Let r be the distance from (x, y) to the origin. Then which of the following defines $\cos \theta$? (1)
- A. $\frac{r}{y}$
B. $\frac{y}{r}$
C. $\frac{r}{x}$
D. $\frac{x}{r}$
-
48. Which one of the following is the solution set of $5^x = 11.2$? [Use $\log 5 = 0.6990$ and $\log 1.12 = 0.0485$] (1)
- A. 1.12
B. 1.5
C. 1.2
D. 1.69
-
49. A boy and girl are both standing on a sun light. The boy's height is 1.6m and casts a shadow of 2m. If the girl's height casts a shadow of 1.8m, then what is the height (in meter) of the girl? (1)
- A. 1.5
B. 1.54
C. 1.68
D. 1.44
-
50. If two vectors $\vec{V} = \vec{AB}$, and $\vec{U} = \vec{BC}$ are given as shown in the figure below . Then what is the vector $\vec{V} + \vec{U}$? (1)
- A. $\vec{AC}$
B. $\vec{BA}$
C. $\vec{CA}$
D. $\vec{CB}$

— END OF EXAMINATION —